

9 NO NEW PETROSTATES

I was eight years old, holding my mother's hand, and walking in the middle of a usually busy highway. There was no car in sight. The silence was eerie but I liked it. I felt glee at the idea that I was allowed to walk on asphalt that I knew was strictly forbidden and deadly under normal circumstances.

But deep down I realized that somewhere, in a country I didn't know, sat scary people I didn't know who had caused this. Who had switched off our normal way of life. And the adults tried to put on a brave face but they were scared.

That was 1973. ([Schramm, 2023](#)) Just an embargo. A few weeks of inconvenience. Cars banned on Sundays. Lines at petrol stations. And then things went back to normal.

For years, I thought of 1973 as our victimhood: the West held hostage by Arab oil producers. I was seeing half the picture. The fossil fuel system doesn't just make us vulnerable. It does something darker.

9.1 The Tyrant's Tool

My parents had lived through the Dutch hunger winter. They didn't throw away food. They told me to clear my plate. "The children in the third world would be very happy with this food," they would say.

Years later, I found a demographic chart with two strange dips. Enormous gaps in population that I couldn't explain at first. The first was the Holodomor, Stalin's deliberate starvation of Ukraine in the 1930s. ([Rudnytskyi, 2015](#)) When Ukrainian peasants resisted collectivization, Stalin sealed the borders, confiscated food, and waited. Approximately four million dead. ([Rudnytskyi, 2015](#)) The second was Mao's Great Leap Forward. Thirty million dead. ([Smil, 1999](#)) I didn't sleep well that week.

Just like the Dutch hunger winter, these weren't famines, I realized. They were weapons. Centralized control of what people need to survive is the tyrant's oldest tool. Energy is now as essential as food. Without it, the lights go out. Communication stops. Transport stops. Buildings go cold. Hospitals fail. Within weeks, cities become uninhabitable.

And like Soviet grain production, fossil fuel infrastructure is easy to control. A few dozen refineries. A handful of pipelines. A few power plants per region. Every one is a chokepoint.

In 2022 Russia throttled gas pipelines to Europe. ([EIA, 2022](#)) Nord Stream 1 deliveries cut to 20%, then halted entirely. ([EIA, 2022](#)) European gas prices reached ten times their historical average. ([Gil Tertre, 2023](#)) Factories closed. Governments scrambled before winter. Putin thought he had Europeans on a leash.

Venezuela's corrupt regime first survived economic collapse because it controls oil revenue. ([TNI, n.d.](#)) ([El País, 2024](#)) Many suspect the coup by American president Trump

against the illegitimate sitting president of Venezuela Maduro in 2025 is also about access to oil. ([Time, 2025](#)) ([BBC, 2025](#)) Saudi Arabia's absolute monarchy distributes energy wealth as patronage. ([Freedom House, 2024](#)) We have built our civilization on a tyrant's tool.

9.2 What Our Oil Money Bought

Fossil fuels concentrate wealth for whoever controls it, often to the detriment of regular people. The pattern is consistent enough to have a name: the resource curse. We are part of that pattern.

In 1953, Iran had a democratically elected prime minister. Mohammad Mossadegh wanted to nationalize Iranian oil, then controlled by the Anglo-Iranian Oil Company, now BP. ([Abrahamian, 2021](#)) As historian Ervand Abrahamian put it: "If nationalization in Iran of oil was successful, this would set a terrible example to other countries where U.S. oil interests were present." ([Abrahamian, 2021](#)) So the CIA and MI6 organized a coup. Operation Ajax. Operation Boot. ([PBS, 2022](#)) Mossadegh was arrested. The Shah took power. The Shah ruled for twenty-six years. ([PBS, 2022](#)) His secret police tortured dissidents. His corruption alienated the middle class. His Western backing alienated the religious establishment. In 1979, the revolution came. ([PBS, 2022](#)) We wanted cheap oil. We got Khomeini.

The House of Saud controls the world's second-largest oil reserves, after Venezuela, and more importantly the cheapest to pump up. ([Worldometer, 2016](#)) Western nations have armed and protected the regime for decades in exchange for reliable oil supplies. ([Khaleel, 2019](#)) The regime exported more than oil. It also exported Wahhabism: the most intolerant form of Islam. ([Monique, 2013](#)) Our money, used by the House of Saud, has built Wahhabi mosques and madrassas across the world. One of my closest friends from university is Dutch. He met his Indonesian wife during his master thesis research and married into a Christian family on Bali. For years, he says, Indonesia was a model of tolerance. The culture was easygoing. His wife's family had Muslim neighbors they'd known for generations. The national motto was "Unity in Diversity." Last time we spoke, he was worried. Saudi-funded mosques have been going up across the archipelago. The Islam they teach is not the relaxed Indonesian tradition. It's imported. Rigid. His wife's family feels under siege. The mosques keep going up.

Fifteen of the nineteen 9/11 hijackers were Saudi nationals. ([9/11 Commission, 2004](#)) The ideology that motivated them was Saudi-funded. The madrassas where they learned were Saudi-built. None were from Afghanistan, where we launched a war. None were from Iraq, where we launched another. ([9/11 Memorial, n.d.](#)) We kept buying their oil.

General Wesley Clark, former NATO Supreme Allied Commander, said in an interview: "The truth about the Middle East is, had there been no oil there, it would be like Africa. Nobody is threatening to intervene in Africa." ([Clark, 2007](#)) Alan Greenspan, Federal Reserve Chair for nearly two decades, wrote in his memoir: "I am saddened that it is politically inconvenient to acknowledge what everyone knows: the Iraq war is largely about oil." ([Greenspan, 2007](#))

Senator John McCain called Russia "a gas station masquerading as a country." (McCain, 2014) Russian oligarchy is built on fossil fuel wealth. That wealth funds the military that invaded Ukraine. Every cubic meter of Russian gas burned in German homes, every oil drum bought globally, helps pay for missiles falling on Ukrainian hospitals.

Then there are those who didn't ask for oil to be under their land. The Standing Rock Sioux, fighting a pipeline through their water supply. The Dakota Access Pipeline was originally routed near Bismarck, the state capital, but moved away due to concerns about contaminating the city's water. (EELP, 2025) The same threat was deemed acceptable for the Indigenous reservation downstream. Standing Rock Tribal Chairman Dave Archambault II told the UN Human Rights Council: "The oil companies and the government of the United States have failed to respect our sovereign rights." (Archambault, 2016) Communities in the Niger Delta live on land contaminated by 6,800 documented oil spills. (UNDP, 2006) Life expectancy there has dropped to 41 years, twenty years below the national average. (BSOEC, 2023) Indigenous peoples of the Amazon watch their forest burned.

We built this system. We funded it. Then we looked away.

9.3 Solar Liberation

Are we trading one vulnerability for another? Will there be solar sheikhs? Fortunately not. Physics won't allow it. Fossil fuels are geologically concentrated. Some countries have oil; most don't. This is why petrostates exist. They exploit this wealth with a few oil wells and refineries. Seize those facilities, and you control the wealth.

Oil is like the spice of the Dune saga. "The spice must flow." Or civilization crumbles. Your car needs new fuel every week. The electricity system needs it constantly. Cut the flow, and everything stops within days. Countries try to stock up but the flows are so massive that they cannot outlast a siege of more than a few weeks, maybe months. That's why OPEC's embargo worked. That's why Russia's gas cutoff caused panic.

Sunlight is different. It falls on every country. Every rooftop. You cannot embargo sunlight. You cannot form a cartel of countries that have sun. Everyone has sun. A solar-powered nation has millions of rooftops, thousands of solar farms. There is no central facility to seize. No chokepoint. No lever.

Solar panels are a stock you build up. Once installed, they keep generating whether or not new panels arrive. Your roof keeps producing. Your battery keeps storing. If supply chains are disrupted, your existing capacity keeps working. You're not dependent on continuous flow. You're standing on accumulated capacity. Oil dependence is flow dependence: continuous vulnerability. Solar is stock dependence: accumulated resilience.

I'm not exactly a prepper. But my home could deliver all the energy I need if the grid went down. My wife isn't a prepper either. But she has ordered me to find a battery system that supports islanding, so the solar panels keep working when the grid fails. Solar isn't just for environmental idealists. It's for anyone who prefers not to be too dependent on systems they don't control.

OPEC has maintained pricing power for fifty years because oil has high margins. Extract it for ten dollars, sell it for seventy. (CNBC, 2024) That margin funds kingdoms and wars. Solar panels are commodity products. Intense competition. Thin margins. (JinkoSolar, 2025) You cannot form OPEC for solar panels. The economics don't support it. There's no cartel premium to capture.

The petrostate model requires geographic concentration, centralized infrastructure, continuous flow dependence, and high margins. Solar breaks all four.

9.4 But What About China?

China dominates clean energy manufacturing. There's no use denying it. Solar panels: about 80% of global production. (IEA, 2022) Polysilicon, the purified silicon that goes into those panels: roughly 90%. (Bernreuter, 2025) Solar wafers: 97%. (IEA-PVPS, 2025) Battery cells: 77%. (Mordor Intelligence, 2024) Electric vehicles: over 70% of global production and rising. (IEA, 2025) These are not small advantages. This is dominance.

And honestly: what China built is an achievement. Not a conspiracy. Not a theft. No abuse of geological advantage. Chinese manufacturers earned their position through strategic investment and deployment at scale while Western countries talked. In 2000, the United States dominated solar manufacturing. (CSIS, 2025) By 2008, Germany funded the global solar boom and paid for the global price reduction. (Wikipedia, 2025) But it was China who invested, scaled, and rode the learning curve to dominance. And there is no resource curse in sight. No country in history has lifted more people out of poverty than China: 800 million people in forty years. (World Bank, 2022)

That said... I like the kind of democracy we have in the Netherlands. I worry about Chinese dominance. About how they treat Uyghurs, Tibetans, and people in Hong Kong and Taiwan. So I understand the questions I hear at every conference and in every comment section. Are our good intentions supporting forced labor? Are we trading one dependence for another? Will oil sheikhs be replaced by solar panel mandarins? OPEC replaced by Beijing?

9.4.1 Forced labor

Polysilicon production has been concentrated in Xinjiang. (IEA, 2022) There are credible reports of coercive labor practices in the region, part of a broader pattern of repression that includes detention camps and forced cultural assimilation for Uyghurs. (OHCHR, 2022) But those who use Uyghur concerns to argue against the energy transition don't seem trustworthy to me. They sound as hollow as those starting to complain about children mining cobalt after we started to use it in electric vehicles instead of gasoline production. And I've noticed that some of the loudest voices on this issue show no interest in Uyghur welfare when the topic isn't solar panels.

It would also be easier to make this moral argument land if the West didn't have a colonialist past. If it hadn't spent a century ignoring labor conditions in its own supply chains. Cobalt mining in the DRC. Garment workers in Bangladesh. Farmworkers in the American South. But that doesn't make forced labor in China ok. So what does help?

Try to buy from supply chains that can prove they don't use forced labor. Buy from verified non-Xinjiang sources. Support traceability requirements. Accelerate manufacturing outside China. The United States has implemented the Uyghur Forced Labor Prevention Act, requiring companies to prove their products are not tainted. ([US Congress, 2021](#)) The EU has adopted a regulation banning products made with forced labor from its market. ([EU, 2024](#)) Major manufacturers have been moving production outside Xinjiang in response. The market is responding. Slowly, imperfectly, but responding. Non-Xinjiang polysilicon capacity is expanding. Traceability systems are being implemented. The pressure is working.

But this is not a uniquely Chinese problem. Moreover, it is not a supply chain security risk that puts us at the mercy of China. Supply security asks whether civilization can function if supply is cut. Conflating them produces moral panic without analytical clarity.

9.4.2 Factories Move, Deposits Don't

Most commentary treats Chinese manufacturing dominance as if it were equivalent to Saudi oil reserves. It is not. The difference is fundamental.

Oil deposits are where plate tectonics and ancient biology put them. Saudi Arabia has 17% of global reserves. ([EIA, 2024](#)) No amount of policy or investment will change that. You cannot decide to have more oil. Geology is permanent.

Manufacturing is where humans put it. The United States dominated solar manufacturing in 2000. Germany dominated in 2008. China dominates in 2024. Each shift happened through investment decisions and industrial policy, not because silicon migrated across the Pacific. What moved twice can move again. And it is moving.

Remember what solar panels and batteries are actually made of. We covered this in the previous chapter. Silicon is made from sand. Sand exists on every continent. High-purity polysilicon requires processing, but processing is manufacturing, not geology. You cannot run out of sand. You can only fail to build processing facilities.

Aluminum is the third most abundant element in Earth's crust. ([Geoscience Australia, 2025](#)) Bauxite deposits span Australia, Guinea, Brazil, Vietnam, Jamaica, India. ([Bauxite Index, 2020](#)) No country has a monopoly. No country can have a monopoly.

Lithium: Australia produces nearly half of global supply, with Chile, China, and Argentina contributing most of the remainder. ([USGS, 2024](#)) Massive deposits have been identified in Bolivia, Germany, the US, Mexico, and across Africa. ([USGS, 2024](#)) The lithium could be mined everywhere. The question is who wants to do it.

LFP batteries, the chemistry that's winning, uses iron and phosphorus next to lithium. Iron is 5% of Earth's crust by weight. ([Visual Capitalist, 2022](#)) Phosphorus has major deposits in Morocco, China, the US, Russia, Jordan. ([USGS, 2022](#)) These are not rare.

China dominates processing and manufacturing. That's undeniable. But China does not have a geological monopoly on any critical material. Manufacturing moves to where

investment goes. That's why Chinese renewables dominance is our choice, while Saudi oil dominance is a given.

9.4.3 Not OPEC 2.0

The petrostate model requires five things: geographic concentration, flow dependence, chokepoint infrastructure, high margins, and non-substitutability. China's position has none of these.

No geographic concentration. Deposits are everywhere and factories can move.

No flow dependence. As we just saw: solar is a stock, not a flow. China could embargo panels tomorrow, and Europe's 338 gigawatts of installed solar keeps generating over 300 terawatt-hours per year. ([European Commission, 2025](#)) Every deployed battery would keep storing. The embargo would affect future installations, not current operations.

No chokepoint infrastructure. Oil wealth concentrates in a few dozen major fields globally. Seize those facilities, and you control the oil wealth. Solar manufacturing is spread across multiple companies and regions, even within China. There is no solar valve to turn off.

No high margins. OPEC maintains pricing power because oil has high margins. Extract for ten dollars, sell for seventy. That margin funds kingdoms and wars. Solar panel gross margins run 10% to 15%. ([JinkoSolar, 2025](#)) Battery margins are still 20-25% for market leaders like CATL, but as batteries commoditize like solar cells, these margins will come down too. ([CnEVPost, 2024](#)) Chinese companies compete ferociously with each other. You cannot form OPEC for solar panels because there's no cartel premium to capture. Coordination requires something to coordinate over. With commodity margins, there's nothing.

Great substitutability. If Chinese panels become unavailable, Indian, Vietnamese, American panels exist. Production can restart or expand elsewhere. The knowledge is not secret. The technology is not proprietary. Try substituting Saudi oil with... what, exactly? Other oil from other places with the same dependencies?

9.4.4 Diversification Is Already Happening

China didn't have solar manufacturing dominance in 2005. They built it through investment and policy over fifteen to twenty years. The same path is available to others and could be sped up.

In the United States, the Inflation Reduction Act has triggered \$369 billion in clean energy investment. ([Treasury, 2022](#)) Solar manufacturing capacity announcements have jumped from roughly 5 gigawatts in 2022 to over 50 gigawatts expected by 2026. ([DOE, 2024](#)) Battery manufacturing announcements exceed 700 gigawatt-hours. ([Bordoff, 2023](#)) First Solar, Qcells, and others are building American facilities. ([Power Magazine, 2025](#)) The IRA is industrial policy. It's what China did, arriving two decades later.

Europe moves more slowly, but the European Solar PV Industry Alliance has set targets. The European Battery Alliance is building manufacturing. Meyer Burger, Enel, and others are building European capacity. The EU Net Zero Industry Act passed in 2024. ([EU Council, 2024](#))

India is running the same playbook China ran. Production-linked incentives for solar manufacturing. Rapidly expanding domestic capacity. Tata, Adani, Reliance entering the market.

Vietnam, Malaysia, and Thailand are already significant manufacturers. Some facilities are Chinese-owned, but they're local, with local workers, diversifying the geography of supply.

Diversification takes two to three years for module assembly, and three to five years for significant upstream manufacturing capacity in cells and wafers. ([SEIA, 2022](#)) Not instant, but not generational either. By 2030, non-Chinese capacity will be substantial. The question is not whether diversification is possible. It's whether we do it.

9.4.5 Chinese are (Primarily) Doing it for Themselves

Something often missing from Western analysis: China has its own reasons to want clean energy that have nothing to do with leverage over the West.

Clean energy is roughly 10% of Chinese GDP and growing. ([Carbon Brief, 2025](#)) Millions of jobs depend on these exports. An export ban would devastate Chinese manufacturers before it hurt Western deployment. China needs customers. An embargo hurts the embargoer.

China has a domestic air pollution crisis. More than a million premature deaths annually from particulate matter. ([Health Effects Institute, 2020](#)) Clean air is a priority for both the government and the population.

China imports more oil than any other country. ([EIA, 2025](#)) Energy security is an obsession for Chinese planners. A healthy obsession, one we might want to copy. The Chinese government understands that solar and wind reduce that vulnerability.

China installed 216 gigawatts of solar in 2023 alone. ([pv magazine, 2024](#)) More than the rest of the world combined. ([IEA, 2024](#)) They added 37 gigawatts of battery storage in 2024 alone, more than doubling their capacity in a single year. ([China Electricity Council, 2025](#)) They are not just doing this for us but primarily for themselves.

When we see Chinese solar factories as a threat, we miss that they're primarily serving Chinese energy needs and Chinese employment. This is an export product. Not a resource we can't do without. Not a weapons arsenal.

9.4.6 The Improbable Lose-Lose Scenario

Imagine China invades Taiwan and announces a complete export ban on all clean energy equipment, effective immediately. Set aside the invasion itself, even though that would clearly be the bigger issue.

Day one through thirty: all existing solar panels keep generating. All existing batteries keep working. All existing wind turbines keep spinning. No immediate impact on current energy supply. Some panic in markets. Some prices spikes. 99% of people notice nothing personally. We still drive on Sunday. Nobody shows pictures of a world burning up.

Months one through twelve: new installations slow dramatically. Prices for available equipment spike two to five times. Non-Chinese manufacturers ramp production to meet backlog demand. Used equipment markets emerge. Some projects get delayed, some cancelled. Some people lose their jobs. But not enough to cause an economic crisis and everybody understands this is a temporary nuisance, not a threat to our way of life.

Years one through three: alternative supply chains expand rapidly. The investment case is now obvious. India, Southeast Asia, the US, and Europe accelerate capacity. Learning curves continue. Technology doesn't forget. Engineers don't forget. Prices normalize as new capacity comes online. People get rehired. The transition continues. Just slower and a bit more expensive.

Year three onward: non-Chinese supply chains mature. Possibly more diversified and resilient than before. Local suppliers flourish. The embargo accelerated what was happening anyway.

Meanwhile, in China: millions of manufacturing jobs lost. Export revenue collapses. Overcapacity. Political backlash domestically.

For us it would be inconvenient and expensive, but not civilization threatening and it would not stop the transition. China's economy would take the biggest hit. Both sides lose, but China would lose most. If it ever came to blows between China and those buying her renewable products, chances are others would punish China with an embargo, not the other way round.

9.5 Power to the People

For a century, energy has meant dependence. On petrostates. On centralized infrastructure. On chokepoints that could be weaponized against us.

Solar and wind are structurally different. When you put solar on your roof, no embargo can touch you. No pipeline dispute can raise your heating bill. When your community has its own generation and storage, the weapon that Stalin used with grain doesn't work.

This book is called "Power to the People." I mean that literally. For the first time in history, the power that runs civilization can be generated everywhere, by everyone. Not concentrated in the hands of whoever won the geological lottery. Not vulnerable to the ambitions of dictators or the calculations of cartels.

There will be no solar-states. The physics won't allow it. We are not trading one master for another. We are stepping outside the master-servant relationship. In terms of energy, we become uncoercible.

So that's the good news about the raw materials we need and how much control that gives others over us. But what about pollution? This was the stuff that goes in. What about the stuff that comes out?